Alexandra McDaniel

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PERSONAL PROFILE

Graduate student in electrical engineering with a strong sense of initiative, a collaborative spirit, and enthusiasm for scientific discovery. Background research in power quality analysis and underwater acoustics. Looking to begin my career in the next 9 months.

EDUCATION

Master's in Electrical Engineering | Laramie, WY — Expected Graduation: May 2026
University of Wyoming
GPA 4.0

Bachelor's in Applied Physics | Provo, UT — July 2024
Brigham Young University
GPA 3.75

Rancho High School | Las Vegas, NV — June 2018
Clark County School District
GPA 4.8 (weighted) | Valedictorian

WORK EXPERIENCE

Graduate Research Assistant | Laramie, WY — August 2024 – Present
University of Wyoming Dept. of Electrical Engineering
- Teaching lab classes (basic circuits and communication theory)
- Analyzed power quality disturbances and the effect they have on a transmission system with renewable energy connections

Independent Research Contractor | Remote — August 2023 – Present
Brigham Young University Hydroacoustic Lab
- Trained deep learning models for seabed classification using sound signatures from merchant ships
- Analyzed the seabed classification accuracy of a robust deep learning model that accounts for global ocean sound speeds when synthetic data from a single area is used during transfer learning

Knobles Scientific and Analysis
- Found evidence of deep sediment heterogeneity in the New England Mudpatch by performing statistical inference using 187 merchant ship noise datasets

** Funded by the Office of Naval Research*

Field Technician | Laramie, WY — May – August 2024
University of Wyoming Dept. of Botany
- Prepared sensors and data acquisition loggers for future deployment in the field
  - Editing data acquisition code in CRBasic
  - Modifying wiring of sensors, basic soldering, calibration, and testing
- Assisted with field surveys of vegetation and water potential in plants

Undergraduate Research Assistant | Provo, UT — January 2022 – July 2023
Brigham Young University Dept. of Physics and Astronomy
- Characterized a laboratory water tank for sound measurements in a model ocean environment
- Introduced temperature and sound speed variability to a laboratory tank to better model the sound speed environments found in the shallow ocean

Physics Lab Manager | Provo, UT January 2022 – December 2022
Brigham Young University Dept. of Physics and Astronomy

- Management experience adhering to SCRUM and AGILE methodology
- Met one-on-one with team members each week to help them make progress on projects
- Created the lab set up schedule and assignments every week
- Improved coordination between physics professors to organize walk-in physics labs for seven classes
- Led organization improvement projects for lab set ups and training documentation

Physics Lab Technician | Provo, UT January – September 2019, May– December 2021
Brigham Young University Dept. of Physics and Astronomy

- Setup, test, and trouble shoot equipment for seven undergraduate physics classes
- Designed and implemented an MRI lab for an introductory lab class for pre-medicine students
- Revised a mechanical work lab for an introductory lab class to implement greater use of LoggerPro and curve fitting techniques
- Maintaining and making small repairs to physics equipment used for lab classes

Lab Teaching Assistant | Provo, UT September 2021 – December 2021
Brigham Young University Dept. of Physics and Astronomy

- Taught a two-hour lab once a week
- Helped students trouble shoot equipment and understand correct experimental practices
- Graded lab reports

FIELD EXPERIENCE

Seabed Characterization Experiment 2022 Falmouth, MA
Student Scientist May 2022

- Assisted with temperature and acoustics device deployment aboard the RV Neil Armstrong
- Processed the ocean temperature data and determined the sound speed of the water column
- The data I helped collect during this month-long experiment is what I have used for underwater acoustics research at both Knobles Scientific and Analysis and Brigham Young University

SKILLS

- Coding languages:
  - Advanced: Python, MATLAB
  - Proficient: MATLAB Simulink
  - Intermediate: LabVIEW, Mathematica
  - Beginner: C++, CRBasic, R
- Software:
  - Proficient: RT-Lab
  - Beginner: SolidWorks, Autodesk Inventor
- Management Experience:
  - Implementation of SCRUM and AGILE methodologies
  - Scheduling experience
  - Interviewing/hiring experience
  - Weekly one-on-ones with team members
  - Implementation of workflow tools such as Click-up and Trello
- Bilingual: English / Spanish

PUBLICATIONS

- Hopps-McDaniel, Alexandra M., et al. "Deep sediment heterogeneity inferred using very low-frequency features from merchant ships." The Journal of the Acoustical Society of America 156.4 (2024): 2265-2274.

- Hopps-McDaniel, Alexandra M., and Tracianne B. Neilsen. "Temperature-induced sound speed variability in a laboratory water tank." Proceedings of Meetings on Acoustics. Vol. 51. No. 1. AIP Publishing, 2023.

- Harmer, Madeline, et al. "Leveraging scientific modeling to engage pre-med undergraduates in physics lab courses." Physical Review Physics Education Research 20.2 (2024): 020150.

- Undergraduate Capstone Project: Hopps, Alexandra, "Characterizing the sound speed variability in laboratory water tank," Advisor: Traci Neilsen (Capstone, Jun 2023).

PRESENTATIONS AT CONFERENCES

- May 2025 - Acoustical Society of America (New Orleans, Louisiana)
  - Presented using deep learning to classify seabed using recordings of ship noise
  - Focused on the need to train robust seabed classification models in the face of sound speed variations in the ocean

- May 2024 - Acoustical Society of America (Ottawa, Canada)
  - Presented on deep sediment layer heterogeneity along the New England continental shelf inferred from acoustic features from passing merchant ships

- March 2024 – Seabed Characterization Experiment Workshop (Kingston, Rhode Island)
  - Poster on deep sediment layer heterogeneity along the New England continental shelf

- May 2023 - Acoustical Society of America (Chicago, Illinois)
  - Presented on introducing sound speed variability in a laboratory water tank for testing the robustness of machine learning algorithms

- February 2023 - Brigham Young University Student Research Conference (Provo, Utah)
  - Presented research involving variability in a laboratory water tank

- January 2023 - Conference for Undergraduate Women in Physics (Santa Cruz, California)
  - Poster about characterizing a laboratory water tank for underwater acoustics measurements
  - Awarded for best poster design

- July 2022 - American Association for Physics Teachers (Grand Rapids, Michigan)
  - Presented on the design of a modeled MRI machine experiment I designed for an undergraduate lab class at Brigham Young University

AWARDS

- 2023: University of Wyoming full graduate assistantship in electrical engineering
- 2022: McKay Family Foundation - BYU Women in STEM Full Tuition Scholarship
- 2018: Brigham Young University Full Tuition Scholarship
- 2018: Valedictorian (class of 850 students)
- 2018: Career and Technical Education Award in Aerospace Engineering
- 2018: Second place in Chemistry Lab at the Nevada State Science Olympiad Competition
- 2017: First place in Public Health at the Nevada State Health Occupation Students of America competition
- 2016: Black Belt in Kenpo Karate
  - Placed in the Annual Las Vegas Tournament in 2008, 2009, 2014, and 2016

RELEVANT COURSE WORK

- Engineering: Cyber Physical Systems, Power Reliability, Power Quality, Power Engineering, Statistical Signal Processing, Communication Theory, Thermodynamics, Biomedical Engineering
- Mathematics: Linear Algebra, Calculus 3, Ordinary Differential Equations, Partial Differential Equations, Theory of Statistics 1 & 2
- Programming: Convolutional Neural Networks, Python lab, MATLAB lab, C++
- Physics: Electricity and Magnetism 1 & 2, Acoustics, Mechanics, Optics, Modern Physics, Intro to Physics 1 & 2
- TA courses: Basic Circuits Lab, Communication Theory Lab, Introductory Physics Lab

UNDERGRADUATE RESEARCH PROJECTS

Three semesters of student-designing experimentation in advanced lab classes

- Built a laser microphone using principles of optics, interferometry, and a transimpedance photodiode to record and analyze songs, chirps, and white noise
- Designed a vacuum system for creation of plasma and plotting the Paschen curve of various gases
- Conducted an experiment to determine the critical temperature of a superconductor sample BiPbSrCaCuO
- Modeled how linear density affected resonance patterns in a Chladni plate

VOLUNTEER EXPERIENCE

Victim Assistance Mentor | Provo, UT — June 2021 – December 2022

Utah Children's Justice Center

- Completed a 40-hour training on sexual violence and child abuse
- Spent 2-3 hours every week doing a fun activity with my mentee

Religious Missionary | Salem, OR — September 2019 – March 2021

Church of Jesus Christ of Latter-day Saints

- Served 9 hours a day for 6 days a week. Taught people from the scriptures and did regular service projects
- Learned to speak Spanish fluently and taught English as a second language

Big Brothers Big Sisters Program | Las Vegas, NV — September 2015 – May 2018

Walter Bracken Elementary School

- Spent one hour each week mentoring an assigned elementary little
- We would often work on homework for half of the time and play a game for the other half